

```
Registered 994000 files...
Registered 995000 files...
Registered 996000 files...
Registered 997000 files...
Registered 998000 files...
Registered 999000 files...
Registered 1000000 files...
Registered 1001000 files...
Registered 1002000 files...
Registered 1003000 files...
Registered 1004000 files...
Registered 1005000 files...
Registered 1006000 files...
Registered 1007000 files...
Registered 1008000 files...
Registered 1009000 files...
Registered 1010000 files...
Registration complete: 1010020 files
vboxuser@vbox1:~/CS550/p2p$
```

```
Registered 1000 files from peer_a; indexed file names: 999208
Registered 1000 files from peer_a; indexed file names: 1000208
Registered 1000 files from peer_a; indexed file names: 1001208
Registered 1000 files from peer_a; indexed file names: 1002208
Registered 1000 files from peer_a; indexed file names: 1003208
Registered 1000 files from peer_a; indexed file names: 1004208
Registered 1000 files from peer_a; indexed file names: 1005208
Registered 1000 files from peer_a; indexed file names: 1006208
Registered 1000 files from peer_a; indexed file names: 1007208
Registered 1000 files from peer_a; indexed file names: 1008208
Registered 1000 files from peer_a; indexed file names: 1009208
Registered 1000 files from peer_a; indexed file names: 1010208
Registered 1000 files from peer_a; indexed file names: 1011208
Registered 1000 files from peer_a; indexed file names: 1012208
Registered 1000 files from peer_a; indexed file names: 1013208
Registered 20 files from peer_a; indexed file names: 1013228
z
```

The screenshot displays a macOS desktop environment. In the foreground, a Termius terminal window is open, showing a list of files registered and indexed. To the right, a web browser window displays the Canvas LMS page for 'Programming Assignment 1'. The page shows a 'Missing' status and a score of '100 Points Possible'. Below the assignment title, there is a download link for 'Programming Assignment 1. cs550_pa1.docx'. A mobile app notification is visible in the bottom right corner, showing a reminder for 'Fred ooo' today. The desktop background features a scenic image of a lake and mountains.

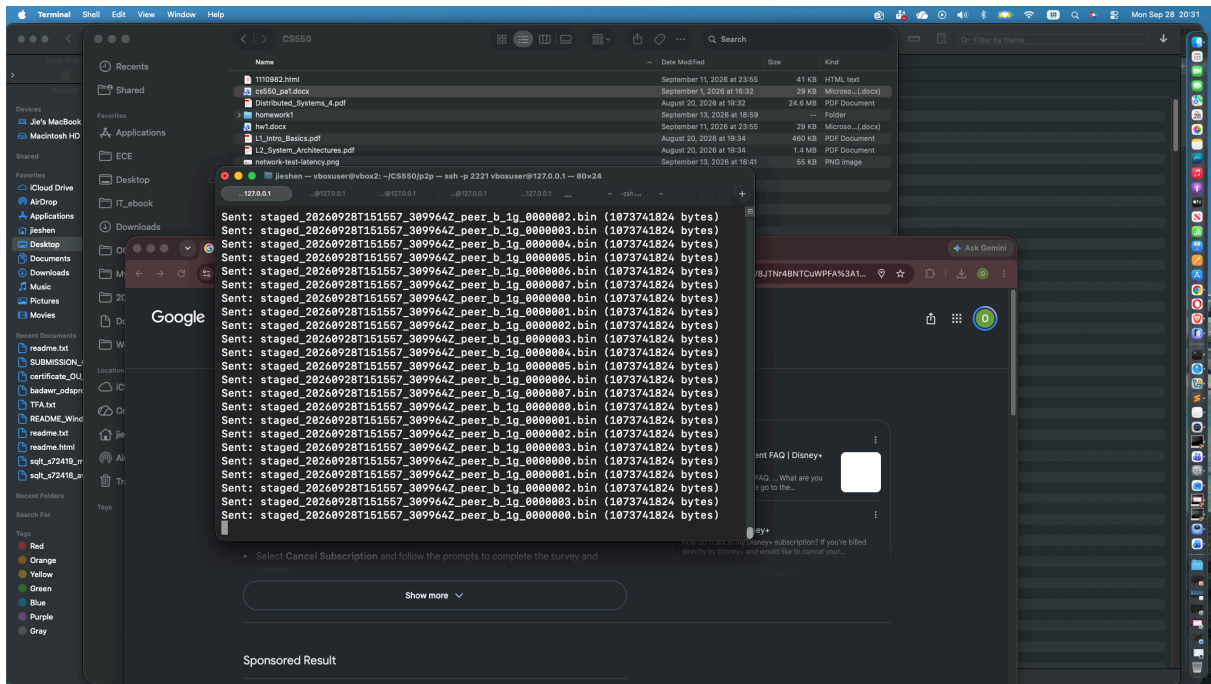
3 Where you will submit
You will have to submit your solution to Canvas as a single compressed package.

4 What you will submit
When you have finished implementing the complete assignment as described above, you should submit your solution to your private git repository. Each program must work correctly and be detailed in-line documented. You should hand in:

1. **Source code:** All of the source code, including proper documentation and formatting.
2. **Makefile/Ant (if needed):** You must use Makefiles or Ant to automate your programming assignment compilation
3. **Output file:** A copy of the output generated by running your program. When it downloads a file, have your program print a message "display file 'foo'" (don't print the actual file contents). When a peer issues a query (lookup) to the indexing server, having your program print the returned results in a nicely formatted manner. Describe any cases for which your program is known not to work correctly
4. **Readme:** A detailed manual describing the structure of your files and directory organization. The manual should be able to instruct users how to run the program step by step. The manual should contain example commands. This should be included as readme.txt in the source code folder.
5. **Report:** A written document (typed, named pa1-report.pdf) describing the overall assignment completion, along with screen shots, tables, figures, and text answering the key questions from the performance evaluation.
6. **Development log:** A log of development activities in an hourly basis. This can be simple but description about what you have done in each active hour. For a team with multiple students, each student should have an individual section in the log. Furthermore, if you are using any AI coding in this assignment, please record all prompts you used in the log.

Submit code/report on Canvas.
Please refer to the late policy in the syllabus.

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Sent: staged_20260928T151557_309964Z_peer_b_1g_0000001.bin (1073741824 bytes)
Sent: staged_20260928T151557_309964Z_peer_b_1g_0000002.bin (1073741824 bytes)
Sent: staged_20260928T151557_309964Z_peer_b_1g_0000003.bin (1073741824 bytes)
Sent: staged_20260928T151557_309964Z_peer_b_1g_0000004.bin (1073741824 bytes)
Sent: staged_20260928T151557_309964Z_peer_b_1g_0000005.bin (1073741824 bytes)
Sent: staged_20260928T151557_309964Z_peer_b_1g_0000006.bin (1073741824 bytes)
Sent: staged_20260928T151557_309964Z_peer_b_1g_0000007.bin (1073741824 bytes)
Sent: staged_20260928T151557_309964Z_peer_b_1g_0000000.bin (1073741824 bytes)
Sent: staged_20260928T151557_309964Z_peer_b_1g_0000001.bin (1073741824 bytes)
Sent: staged_20260928T151557_309964Z_peer_b_1g_0000002.bin (1073741824 bytes)
Sent: staged_20260928T151557_309964Z_peer_b_1g_0000003.bin (1073741824 bytes)
Sent: staged_20260928T151557_309964Z_peer_b_1g_0000000.bin (1073741824 bytes)
Sent: staged_20260928T151557_309964Z_peer_b_1g_0000001.bin (1073741824 bytes)
Sent: staged_20260928T151557_309964Z_peer_b_1g_0000002.bin (1073741824 bytes)
Sent: staged_20260928T151557_309964Z_peer_b_1g_0000003.bin (1073741824 bytes)
Sent: staged_20260928T151557_309964Z_peer_b_1g_0000000.bin (1073741824 bytes)
```

```
vboxuser@vbox1:~/CS550$ ls
p2p protocol.py
vboxuser@vbox1:~/CS550$ cd p2p/
vboxuser@vbox1:~/CS550/p2p$ python3 peer_server.py --config config.json
Peer ID: peer_a
Listening on 0.0.0.0:9001
Shared directory: /home/vboxuser/data/CS550/p2p/peer_a/shared
```